Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/14/2014
PI Name:	Hahn, Inseob Ph.D.		
Project Title:	Investigation of equilibration near the liquid-gas critical point in microgravity utilizing DECLIC/ALI facility		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Fundamental Physics Rapid Turn Around Flight Opportunity
Start Date:	10/01/2010	**End Date:**	09/30/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended for reflight to 9/30/2017 (per PI and U. Israelsson/JPL)-Ed., 5/12/14 NOTE: Extended to 3/31/2014 per PI (Ed., 10/21/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zhong, Fang (Jet Propulsion Laboratory) Beysens, Daniel (Commissariat a l'Energie Atomique at Grenoble and Ecole Supérieure de Physique et Chimie Industrielle) Garrabos, Yves (Centre National de la Recherche Scientifique (CNRS)-)-Institut de Chimie de la Matière Condensee de Bordeaux (ICMCB) Laboratory) Lecoutre, Carole (Centre National de la Recherche Scientifique (CNRS)-Institut de Chimie de la Matière Condensee de Bordeaux (ICMCB) Laboratory)		
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Performance Goal No.:			
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Task Description:	A key feature of a continuous phase transition in many physical systems in Nature is scale-invariance in fluctuations of the order parameter, which implies continuity between large and small scale fluctuations associated with the transition. Scale-invariance occurs in many systems that are still not completely understood, such as the universe at the onset of the big bang as currently envisioned by cosmologists, the clustering of galaxies, massless fields in elementary particle physics, the distribution of earthquakes, and turbulence in fluids and plasmas. In each of these examples there is a wide range of scales over which some phenomena vary as a power-law of the scale. The Nobel Prize winning Renormalization Group theory is the most successful approach thus far developed to derive and explore the relations between the characteristics of a system as viewed on different length scales. Many scientific investigations using simple fluid systems have been performed on the ground. The predictions of the RG theory have been reasonably well validated. Nevertheless, the theory is based on “seemingly innocuous” assumptions that will stand or fall based on more sophisticated experiments. Experimental studies of critical phenomena using simple fluids on Earth have been very difficult to perform due to the gravity induced density gradient. NASA’s MISTE/COEX (microgravity scaling theory experiment/coexistence curve experiment) flight experiments were high precision studies of static critical phenomena planned on board the ISS. Those experiments were to address one of the most difficult experimental challenges in studying static properties of near critical fluid in microgravity, which is associated with the equilibration time of the sample. Recently, the DECLIC (Dispositif pour l'Etude de la Croissance et des LIquides Critiques) facility supported by CNES/NASA was successfully integrated into the International Space Station (ISS). The DECLIC facility developed by CNES investigators allows precision studies of non-equilibrium phenomena near the liquid-gas critical point. Multiple science inserts for the DECLIC facility have been developed to study room temperature and high temperature critical fluids. These inserts are currently on-board ISS to study the boiling phenomena, phase separation and piston effect of near critical fluids. The DECLIC/ALI experiment is to be performed in the near future. Detailed discussions with the DECLIC PI’s have clearly indicated that additional important scientific investigations can be carried out utilizing the available DECLIC/ALI hardware. Taking advantage of the current CNES funded DECLIC facility, we propose to NASA to study the temperature and density relaxation behavior of the fluid (SF6) near its liquid-gas critical point by utilizing DECLIC/ALI experiments. By analyzing experimental data taken under microgravity conditions, critical data to understand the equilibration processes of a near critical fluid, especially in the two-phase region, can be obtained that is not feasible on the Earth. The results of data analysis, modeling, simulations, and ground support experiments will enhance the scientific return of the proposed DECLIC investigations. This study of equilibration near a critical fluid will directly support the objectives of NASA’s MISTE/COEX flight experiment and benefit future critical phenomena studies utilizing the DECLIC facility. The research team of this proposal consists of NASA MISTE/COEX investigators and CNES DECLIC principal investigators. APRIL 2014--ADDITIONAL INFORMATION FOR REFLIGHT PROJECT (ALI-R) A re-flight of the DECLIC Alice-Like-Insert (ALI-R) under microgravity conditions is to measure the light transmission and the turbidity of sulfurhexafluoride (SF6) fluid in the single-phase region. The new experiment will utilize the existing DECLIC facility, the ground control facility at CNES, and a new measurement cell. The ALI-R will be able to experimentally measure the effects of the non-zero value of the Green-Fisher exponent on the liquid-gas critical point for the first time. A measurement of the turbidity of the SF6 sample with a precision of 0.3% at ~ 1mK above the critical temperature could test theoretical predictions. This goal can be achieved using the existing DECLIC hardware. However, the sample cell will need to be refilled at an average density within 0.1% of the critical density. The proposal research team includes the current CNES DECLIC principal investigator. The commitment from the CNES regarding the refurbishment and operation of the experiment is established. The continuing support from NASA for the utilization of the DECLIC facility is planned to study the critical phenomena, recommended by the NRC’s decadal survey report.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will provide a unique opportunity to measure experimental properties of a fluid near its liquid-gas critical point without gravitational effects and to refine the theoretical equation of state models. The critical and supercritical state of fluids exhibit unique properties. For instance, the fluid near the critical point and supercritical state shows very large compressibility. The use of these fluids, for instance, for the storage of cryogenic propellant, under normal gravity and/or reduced gravity conditions raises fundamental questions regarding fluid dynamics, heat transfer, interfacial phenomenon. These experimental works will enhance the understanding of basic physics regarding the fluids near critical state. Potential impacts are fields of applications like supercritical fluid extraction, Dry-cleaning, nano and micro particle formation, supercritical drying, supercritical water oxidation.
Task Progress:	The recent operations of DECLIC-ALI on board ISS demonstrated that the turbidity measurements can take benefit from the high-quality thermal and optical environment of the ALI insert and the microgravity environment of the ISS. The turbidity measurements were performed very close to the critical point in a constant density test cell in the region that is not accesible on ground. We also developed a new crossover master model (CMM) of the parametric equation-of-state for a simple fluid system with no adjustable parameter. The CMM model and the phenomenological CPM model are compared with the turbidity data. The preliminary analysis of the different theoretical formulations of the Ising-like equation-of-state models has shown that the measurements of the singular properties at off-critical density (~1.7 %) can still be an accurate probe for testing the recent classical-to-critical crossover models. As a follow-up experiment, the ALI-R is proposed to measure the effects of the value of the Green-Fisher exponent in the liquid-gas critical point. By analyzing the theoretical predictions, a measurement of the turbidity of the sample SF6, with a precision better than 0.3 % near the critical temperature (~1mK above the Tc) and 0.1 % near the critical density, could test the theoretical predictions. The DECLIC ALI-R will be led by Dr. Y. Garrabos (CNES PI) and Dr. I. Hahn (NASA PI). The plan requests one-time re-flight of the same ALI insert with a newly filled sample cell with the SF6 closer to the critical density. Approximately total one-year flight operation of the DECLIC ALI-R is planned.

Bibliography Type:	Description: (Last Updated: 01/02/2025)
Abstracts for Journals and Proceedings	Lecoutre C, Garrabos Y, Beysens D, Hahn I. "Critical phenomena studies utilizing the ISS DECLIC ALI: what's next?" 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Program and abstracts, p. 61-2. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Lecoutre C, Garrabos Y, Beysens D, Hahn I. "Ising-like equation of state models in the light of DECLIC ALI turbidity measurements in off-critical SF6 on board ISS." 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Program and abstracts, p. 65-6. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Articles in Peer-reviewed Journals	Garrabos Y, Lecoutre C, Marre S, Guillaument R, Beysens D, Hahn I. "Crossover equation of state models applied to the critical behavior of xenon." Journal of Statistical Physics. Submitted as of July 2014. , Jul-2014
Articles in Peer-reviewed Journals	Lecoutre C, Garrabos Y, Beysens D, Nikolayev V, Hahn I. "Boiling phenomena in near-critical SF6 observed in weightlessness." Acta Astronautica. 2014 Jul-Aug;100:22--9. http://dx.doi.org/10.1016/j.actaastro.2014.03.012 , Jul-2014
Papers from Meeting Proceedings	Garrabos Y, Lecoutre C, Beysens D, Nikolayev V, Hahn I. "Boiling phenomena in near-critical SF6 observed in weightlessness." IAC 64: 64th International Astronautical Congress, Beijing, China, September 23-27, 2013. IAC 64: 64th International Astronautical Congress, Beijing, China, September 23-27, 2013. Paper 20124. , Sep-2013